# Ecology: abundance and distribution

*9C Threat from disease · Lesson 6 of 7*

Date _____ Class _____

## Today I can 今天我能

- ☐ I can choose suitable apparatus for measuring the abundance and distribution of organisms.
- ☐ I can use data from a sample to estimate a population size and say how accurate the estimate is.

## What happens today

1. Do Now: counting the uncountable
2. Learn: Abundance and sampling · Estimating population size · How accurate is the estimate? · and more
3. Practical: Quadrat sampling
4. Challenge, then the exit ticket

Student Book pages 48 to 49 · Workbook pages 34 to 35

**What I learned today (one or two sentences) 今天我学到了：**

# Key Vocabulary

| Term | 中文 | Definition |
|---|---|---|
| **abundance** | 丰度 | how common a species is in a habitat |
| **distribution** | 分布 | how organisms are spread out in an area |
| **population** | 种群 | all the organisms of one species in an area |
| **sample** | 样本 | a small selected part of the total number |
| **estimate** | 估算 | an approximate calculation of something |
| **quadrat** | 样方 | a square frame used to sample plants or slow-moving animals |
| **transect** | 样线 | a line along which samples are taken |
| **pitfall trap** | 陷阱 | a container sunk into the ground to catch small ground animals |
| **sweep net** | 捕虫网 | a net swept through long grass to catch insects |
| **random** | 随机的 | chosen by chance, not by the person |
| **biodiversity** | 生物多样性 | the number of different species in an area |

**Command words 指令词:** Calculate 计算 · Complete 完成（填写） · Describe 描述 · Draw 画出 · Estimate 估算 · Give 给出 · Match 连线（配对） · Predict 预测 · Suggest 提出（建议） · Tick 打勾

# Quick Review

*counting the uncountable*

During a disease outbreak, scientists cannot count every infected person, so they estimate.

**1** What does 'estimate' mean?

**2** Suggest how you could estimate the number of dandelions on the school field without counting them all.

**3** Name one abiotic and one biotic factor that could affect how many dandelions there are.

# Core 1 of 5

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**1** Match each word to its definition. Draw a line.

| | |
|---|---|
| abundance | an approximate calculation of something |
| distribution | the number of different species in an area |
| sample | how common a species is in a habitat |
| estimate | a small selected part of the total number |
| biodiversity | how organisms are spread out in an area |

**2** Tick one box in each row to show the best method for sampling each organism.

| Organism | pitfall trap | sweep net | quadrat | pooter |
|---|---|---|---|---|
| ground beetles | | | | |
| insects in long grass | | | | |
| daisies on a lawn | | | | |
| tiny insects on a leaf | | | | |
| limpets on a rocky shore | | | | |

# Core 2 of 5

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**3** Students counted one species of snail in eight quadrats placed at random in a field 30 m by 25 m. Each quadrat was 0.5 m by 0.5 m.

| Quadrat | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Number of snails | 3 | 1 | 0 | 2 | 4 | 1 | 2 | 3 |

**a)** Calculate the total number of snails counted.

**b)** Calculate the total area sampled by the eight quadrats, in m2.

# Core 3 of 5

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**3** *continued*

**c)** Calculate the area of the whole field, in m2.

**d)** Estimate the population of this snail species in the whole field. Show your working.

# Core 4 of 5

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**4** Record your own quadrat results, then estimate the population of your plant in the 10 m by 10 m area.

**a)** Complete the table.

| Quadrat | Coordinates | Number of plants |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |
| total | | |

# Core 5 of 5

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**4** *continued*

**b)** Estimate the population: total x 100 m2 / 2 m2. Show your working.

# Challenge 1 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C1** Ecologists compared the biodiversity of birds in rainforest and in oil palm plantations that replaced some of the forest. They walked along transects through each area, recording every bird they saw or heard.

**a)** Suggest why the scientists used transects rather than quadrats.

**b)** Give two things the scientists should keep the same so that the results from the two areas can be compared fairly.

# Challenge 2 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C1** *continued*

**c)** Oil palm plantations have trees planted in rows, all the same age; rainforest has many kinds of trees of different sizes. Predict in which area the birds would show a more regular distribution, and give a reason.

**C2** There are many types of grass, each with slightly different leaf shapes. Describe how you would find out which of four species was the most abundant in a large field.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** Why do ecologists take samples instead of counting every organism?

**2** 5 daisies were found in one 0.25 m2 quadrat. The lawn is 40 m2. Estimate the daisy population.

**3** Give one reason why quadrats should be placed at random.

**Today I can:** Goal 1 ☐ Goal 2 ☐